METRIC

DOD-S-63544(AR)
5 January 1982

MILITARY SPECIFICATION

SHIM, METRIC,
GENERAL SPECIFICATION FOR

This specification is approved for use by US Army Armament Research and Development Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers metrically dimensioned steel shims.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-H-1581 - Hardware (Fasteners and Related Items), Packaging of

MILITARY

DOD-S-63544/1(AR) - Shim, CRES, Metric

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, ATTN: DRDAR-TST-S, Dover, NJ 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 5365

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STANDARDS

FEDERAL

FED-STD-66 - Steel: Chemical Composition and Hardenability
FED-STD-151 - Metals: Test Methods

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specificaton to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DODISS and the supplement thereto, if applicable.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI B46.1 - Surface Texture

(Applicaton for copies should be addressed to the American National Standards Institute, Incorporated, 1430 Broadway, New York, NY 10018.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ANSI/ASTM A 380 - Cleaning and Descaling Stainless Steel Parts, Equipment, and Systems

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references other than specification sheets cited herein, the text of this specification shall take precedence.

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3. REQUIREMENTS

3.1 Specification sheets. The individual item requirements shall be as specified herein and in accordance with the applicable specification sheet. In the event of any conflict between the requirements of this specification and the specification sheet, the latter shall govern.

3.2 Material. Recycled and reclaimed materials shall be used to the maximum extent practicable.

3.2.1 Corrosion-resistant steel. The shims shall be made from corrosion-resistant steel, composition 300 series conforming to FED-STD-66 and shall have a minimum ultimate tensile strength of 586 MPa.

3.3 Cleaning and descaling. Corrosion resistant steel shims shall be cleaned and descaled in accordance with ASTM A 380.

3.4 Dimensions. Dimensions and tolerances shall be in accordance with the applicable specification sheet (see 3.1) and as specified herein.

3.5 Surface roughness. Shims shall have a maximum surface roughness of 1.6 µm, determined in accordance with ANSI B46.1.

3.6 Cracks. Shims shall be free from cracks, flaws and pits in any location. A crack is a clean crystalline fracture passing through or across the grain boundaries without inclusion of foreign elements (see 4.6.1.1).

3.7 Workmanship. Shims shall be free from surface contamination, tool marks and other imperfections which may adversely affect useability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed reqirements.

4.2 Classification of inspections. The inspections specified herein are classified as follows:

a. Material inspection (see 4.3)
b. Quality conformance inspection (see 4.4)

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4.3 Material inspection. Material inspection shall consist of certification supported by verifying data that the materials used in fabricating the shims, are in accordance with 3.2 and 3.3.

4.4 Quality conformance inspection. Quality conformance inspection shall be as specified in table I.

TABLE I. Quality conformance inspection.

Inspection	Requirement paragraph	Test method paragraph
Dimensions	3.4	4.6.1
Cleaning and descaling	3.3	4.6.2
Tensile strength	3.2.1	4.6.3

4.4.1 Inspection lot. An inspection lot shall consist of all shims covered by a single specification sheet, produced under essentially the same conditions, and offered for inspection at one time.

4.4.2 Rejected lots. If an inspection lot is rejected, the contractor may rework it to correct the defects, or screen out the defective units, and resubmit for inspection. Resubmitted lots shall be inspected using tightened inspection. Such lots shall be separate from new lots, and shall be clearly identified as reinspected lots.

4.4.3 Sampling for visual and dimensional examination. Sampling shall be in accordance with inspection level S-3 of MIL-STD-105 with an acceptable quality level (AQL) of 2.5 percent defective for major defects. For minor defects, inspection level S-3 and an AQL of 4.0 percent defective shall apply.

4.4.4 Sampling for cleaning and descaling test. Sampling for cleaning and descaling test shall be in accordance with inspection level S-2 of MIL-STD-105 with an AQL of 2.5 percent defective.

4.4.5 Sampling for tensile strength test. Sampling for tensile strength test shall be in accordance with inspection level S-1 of MIL-STD-105 with an AQL of 1.5 percent defective.

4.5 Inspection of packaging. The sampling and inspection of the preservation, packing, and container marking shall be in accordance with the requirements of PPP-H-1581.

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4.6 Methods of inspection.

4.6.1 Visual and dimensional examination. Samples taken as specified in 4.4.3 shall be thoroughly examined to determine conformance with this specification and applicable specification sheet. Examination shall be conducted in accordance with table II.

TABLE II. Classification of defects.

Category	Defect	Inspection method
Critical	None defined	
Major		
101	Inside diameter, not as specified (see 3.4)	SIE 1/
102	Outside diameter, not as specified (see 3.4)	SIE
103	Evidence of cracks, flaws or pits (see 3.6)	Visual
104	Thickness, not as specified (see 3.4)	SIE
Minor		
201	Cleaning and descaling missing or incomplete (see 3.3)	Visual
202	Surface roughness, not as specified (see 3.5)	SIE
203	Workmanship (see 3.7)	Visual

1/ Standard inspection equipment

4.6.1.1 Examination for evidence of cracks, flaws or pits. Samples shall be examined for evidence of cracks, flaws or pits using a 3X or 5X magnification visual aid.

4.6.2 Cleaning and descaling. Samples taken as specified in 4.4.4, shall be inspected for adequacy of cleaning and descaling in accordance with the applicable specification of 3.3.

4.6.3 Tensile strength. Samples taken as specified in 4.4.5 shall be tested for tensile strength in accordance with FED-STD-151.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with PPP-H-1581 (see 6.2).

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6. NOTES

6.1 Intended use. Shims acquired under this specification are intended for maintaining a predetermined distance between two surfaces.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number and date of this specification and applicable specification sheet.
b. Applicable specification sheet (see 3.1) part number.
c. Level (degree) of protection in accordance with PPP-H-1581, ordering data (see 5.1).

Custodians:
Army - AR

Review activities:
Army - AV, MI
DLA - IS

User activities:
Army - ME
Navy - OS, SH

Agent:
DLA - IS

Preparing activity:
Army - AR

(Project 5365-A004)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/ vendors who are prospective suppliers of the product are invited to submit comments to the government. *Fold on lines on reverse side, staple in corner, and send to* preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE: DOD-S-63544(AR) Shim, Metric, General Specification for Project 5365-A004

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?
   [ ] YES [ ] NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 1426 REPLACES EDITION OF 1 JAN 68 WHICH MAY BE USED S/N 0102-014-1802

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